

Module 1 Lab 3 — Hands-On Problem 1.2: Choosing an Insurance Provider

Name: [Your Name]

Course: [Course / Section]

GitHub Repo: [link to your portfolio repo]

Problem (paraphrased from the textbook, Hands-On Problem 1.2): An imaginary dataset gives three ratings each (from the three most recent customers) for three auto insurance providers. Using only these ratings, which provider should be chosen, and why?

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
```

1. The Raw Data

In [2]:

```
data = {
    "Insurance Provider": ["GEICO", "GEICO", "GEICO",
                          "Progressive", "Progressive", "Progressive",
                          "USAA", "USAA", "USAA"],
    "Rating": [4.7, 8.3, 9.2,
               7.4, 6.7, 8.9,
               3.8, 6.3, 8.1]
}
ratings_df = pd.DataFrame(data)
print(ratings_df.to_string(index=False))
```

Out:

Insurance Provider	Rating
GEICO	4.7
GEICO	8.3
GEICO	9.2
Progressive	7.4
Progressive	6.7
Progressive	8.9
USAA	3.8
USAA	6.3
USAA	8.1

2. Summary Statistics: Min, Mean, Max per Provider

The average alone can be misleading. The **minimum** rating tells us about the worst experience a customer actually reported.

In [3]:

```
summary = ratings_df.groupby("Insurance Provider")["Rating"].agg(["min", "mean", "max"]).round(2)
summary = summary.reindex(["GEICO", "Progressive", "USAA"])
print(summary.to_string())
```

Out:

	min	mean	max
Insurance Provider			
GEICO	4.7	7.40	9.2
Progressive	6.7	7.67	8.9
USAA	3.8	6.07	8.1

3. Visualizing the Risk Trade-off

Each point below is the **mean** rating; the bars extend down to the **minimum** and up to the **maximum** reported rating, so the length of each bar shows how much the experience varied for that provider.

In [4]:

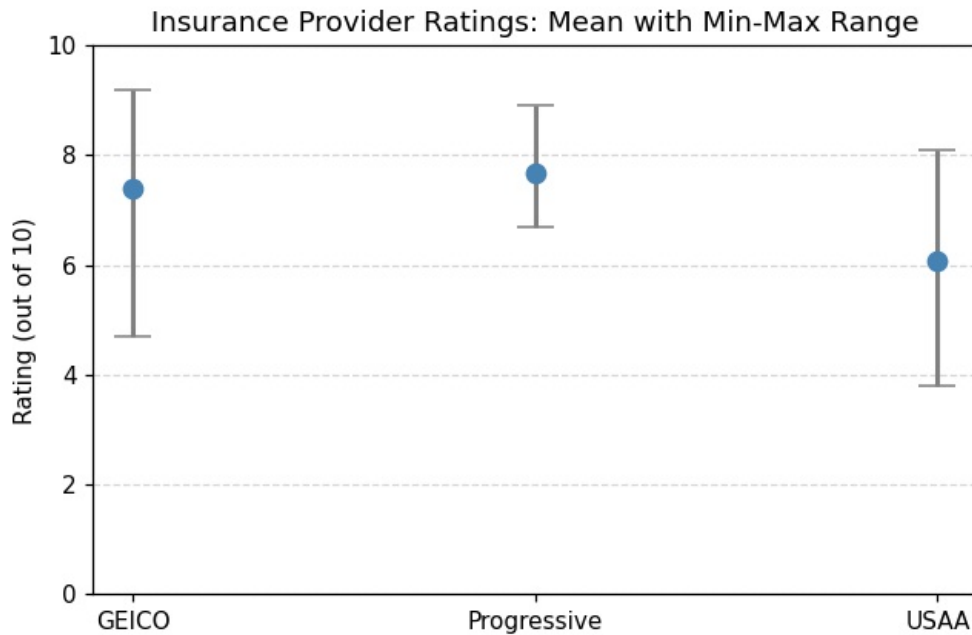
```

providers = summary.index
means = summary["mean"]
mins = summary["min"]
maxs = summary["max"]
lower_err = means - mins
upper_err = maxs - means

fig, ax = plt.subplots(figsize=(6, 4))
ax.errorbar(providers, means, yerr=[lower_err, upper_err], fmt='o', capsize=8,
            markersize=8, color='steelblue', ecolor='gray', elinewidth=2)
ax.set_ylabel("Rating (out of 10)")
ax.set_title("Insurance Provider Ratings: Mean with Min-Max Range")
ax.set_ylim(0, 10)
ax.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()

```

Out:



4. Analysis and Recommendation

(Replace this cell with your own answer, 5–8 sentences. Per the course AI-Use Policy, this reasoning must be your own words. Make sure you: name ONE provider, explicitly discuss the minimum/worst-case rating — not just the mean — and explain the trade-off in plain English, as if to a manager.)

TODO: Write your recommendation here.